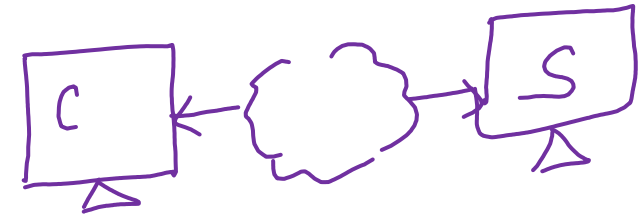
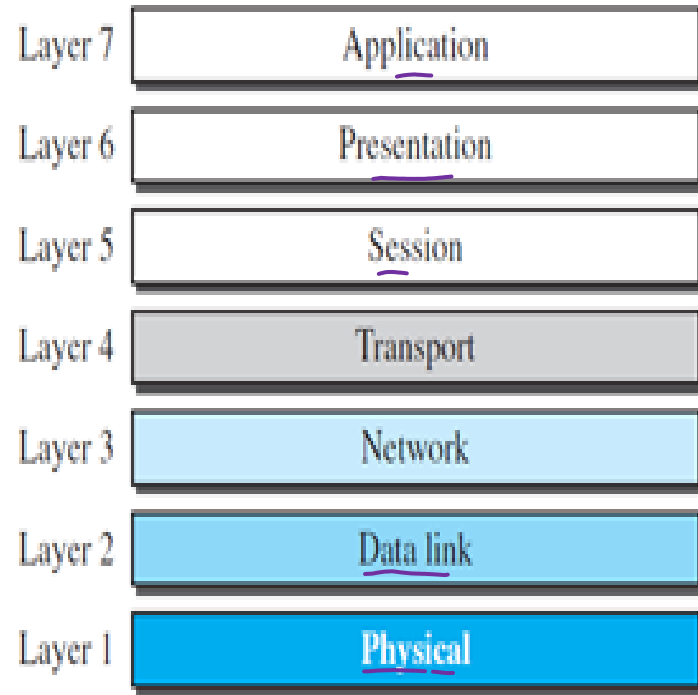


# OSI Layers

-Dr. Sumana Maiti

# OSI Layers

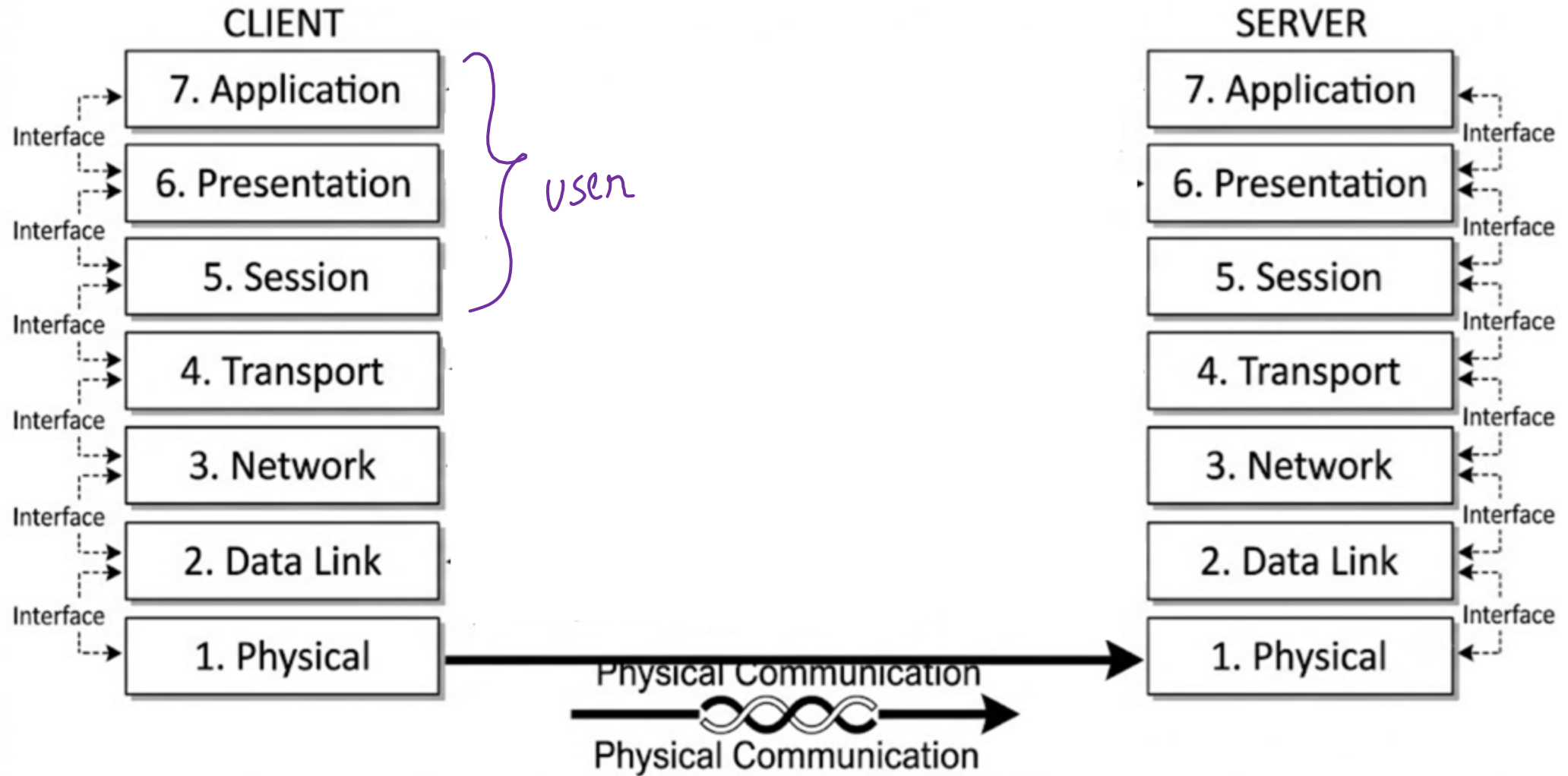


functions

- (1) Error Control
- (2) Flow Control
- (3) Access Control
- (4) Routing
- (5) Encryption/Decryption

# Why Layering

- **Divide and Conquer** - One big problem is divided into multiple smaller problems and solve them independently
- **Encapsulation** - Keep the closely related functions in one layer
- **Abstraction** - One layer need not know the implementation of other layer
- **Testing**



# Application layer

- It provides user interface between user applications (eg. web browser, email client) and underlying network.
- It implements various application level protocols.  
HTTP  
FTP  
SMTP  
IMAP  
POP  
TELNET
- It ensures that destination system or the service can be reached properly.
- It checks whether both the systems have necessary requirements to communicate with each other.

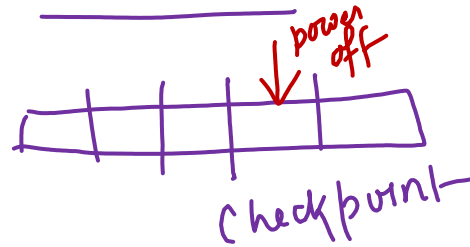
# Presentation Layer

- Character translation: Translates the message
- Encryption/Decryption of the message
- Compression: Zip the data or unzip the data

# Session Layer

- Authentication: Using **Digital signature**.

- Checkpointing

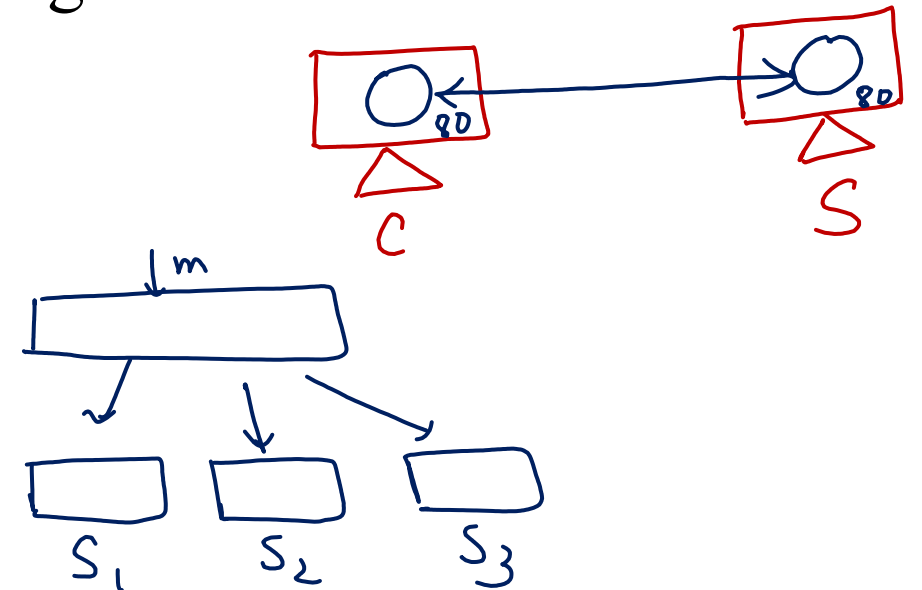
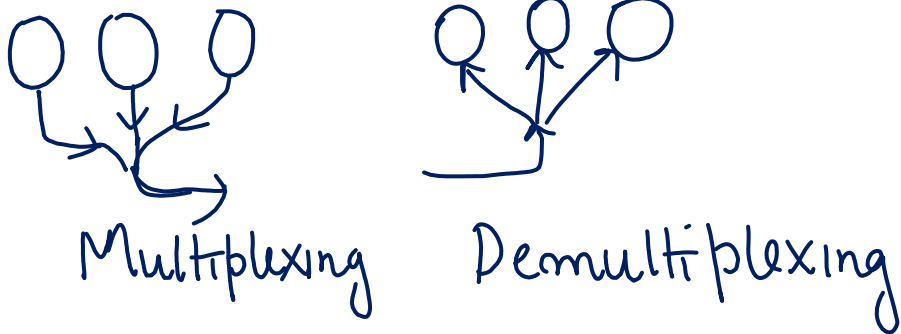


- **Synchronization**: Sometimes audio and video come in different files.
- **Dialog Control**: Even the link is full duplex, we want it to be half duplex.



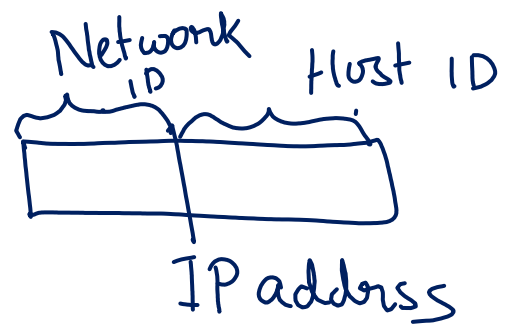
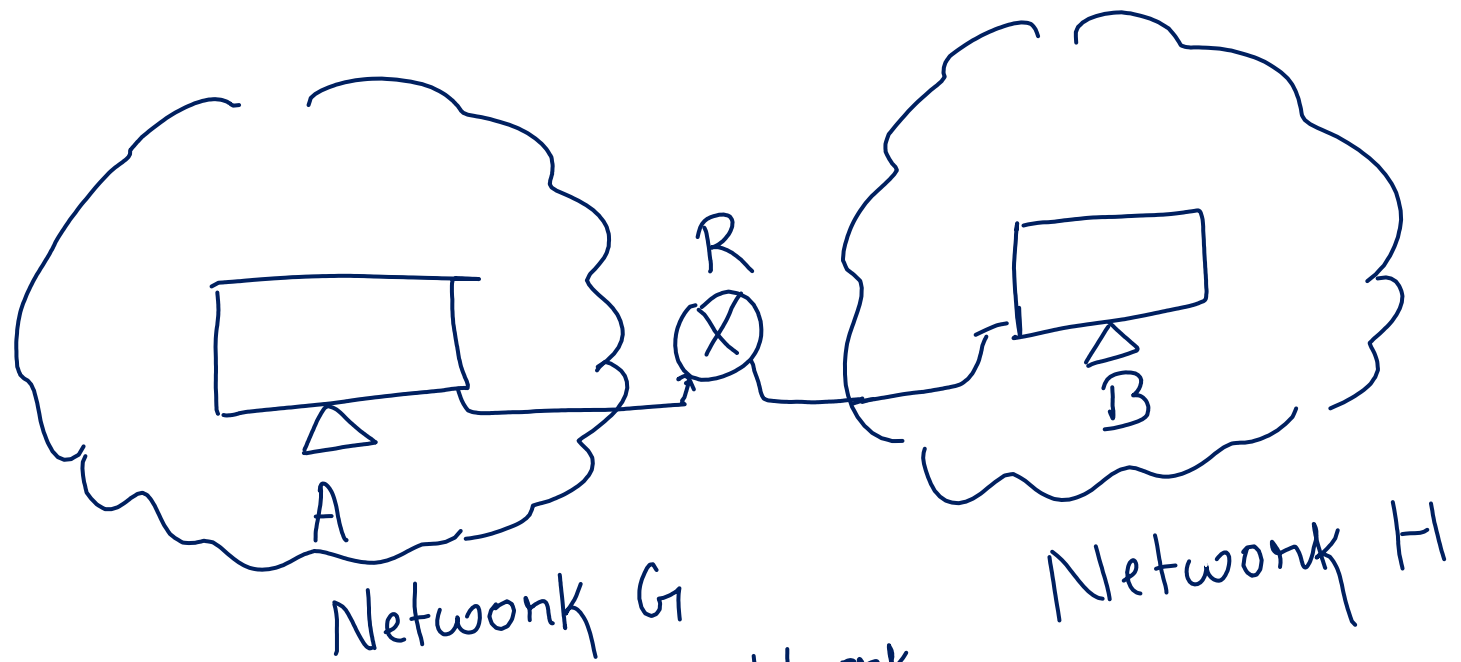
# Transport Layer

- End to End: How message goes from one process to another process
- **Segmentation**: Divides the data into small segments
- **Multiplexing and Demultiplexing**



# Network Layer

- Host to Host Connectivity: Delivers packet **between source host and destination host**
- Logical Addressing: **Unique numerical identifier** assigned to every device connected to a network *IP address*
- Routing: To take the decision **in which network you want to send the packet**, routing table is needed.

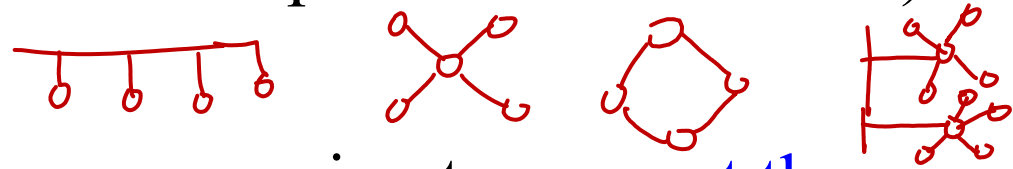


# Data Link Layer

- Flow Control: Sender should transmit packets in such a way that receiver's buffer does not overflow. *Stop and wait, Go back N, Selective Repeat*
- Error Control: (i) Entire packet is loss, (ii) Receiver receives the packets, but inside the packet, some bits are corrupted. *CRC, Checksum*
- Access Control: If many stations are accessing the same channel, more than one channel should not send the data at the same time.
- Physical addressing: *CSMA/CD* Unique within network. 48 bits number (Vendor id, *collision* *SD* *id*)  
*date of manufacturer, serial number*

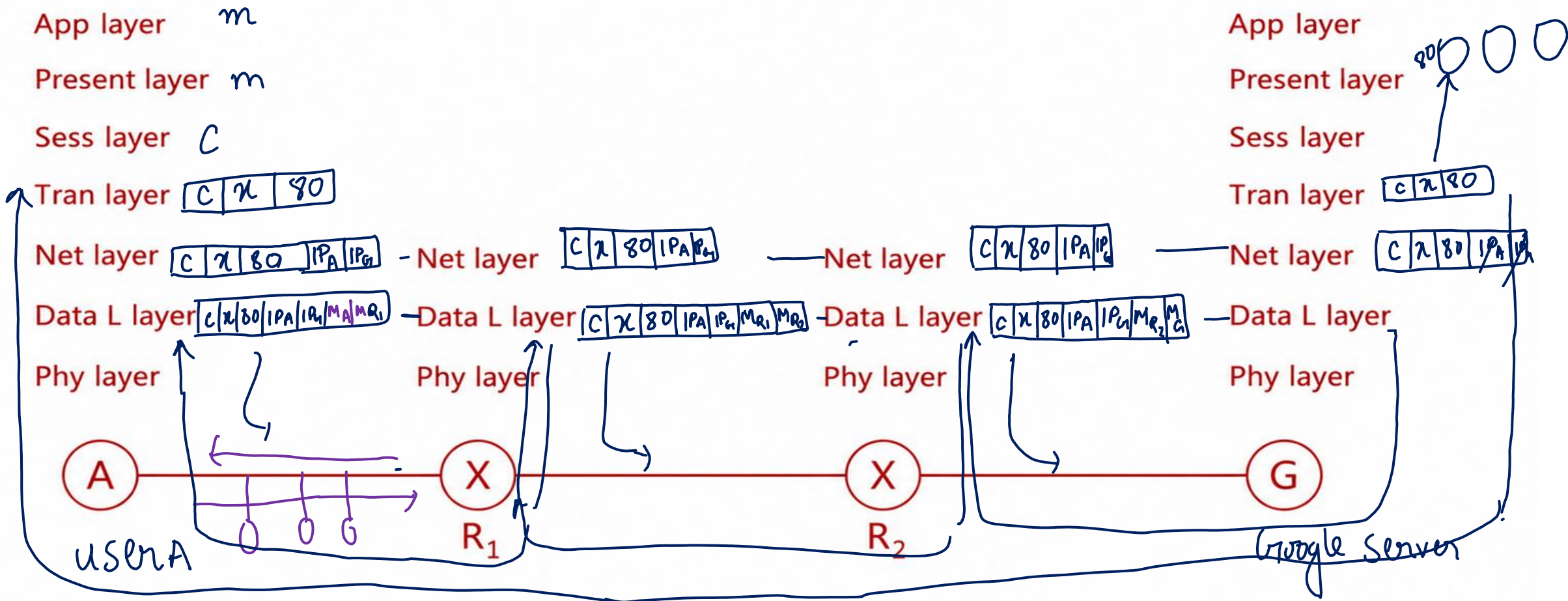
# Physical Layer

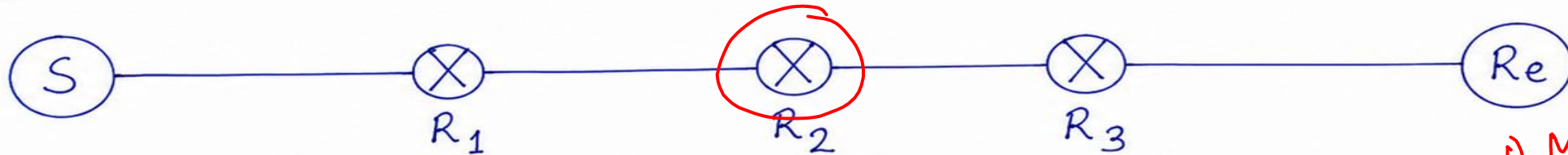
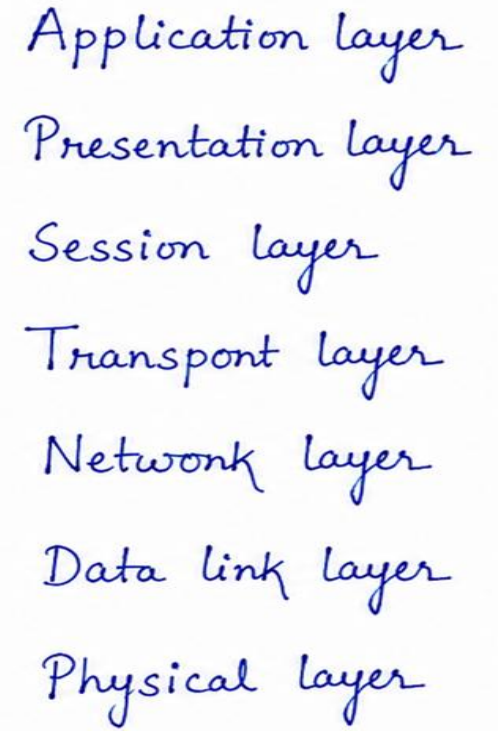
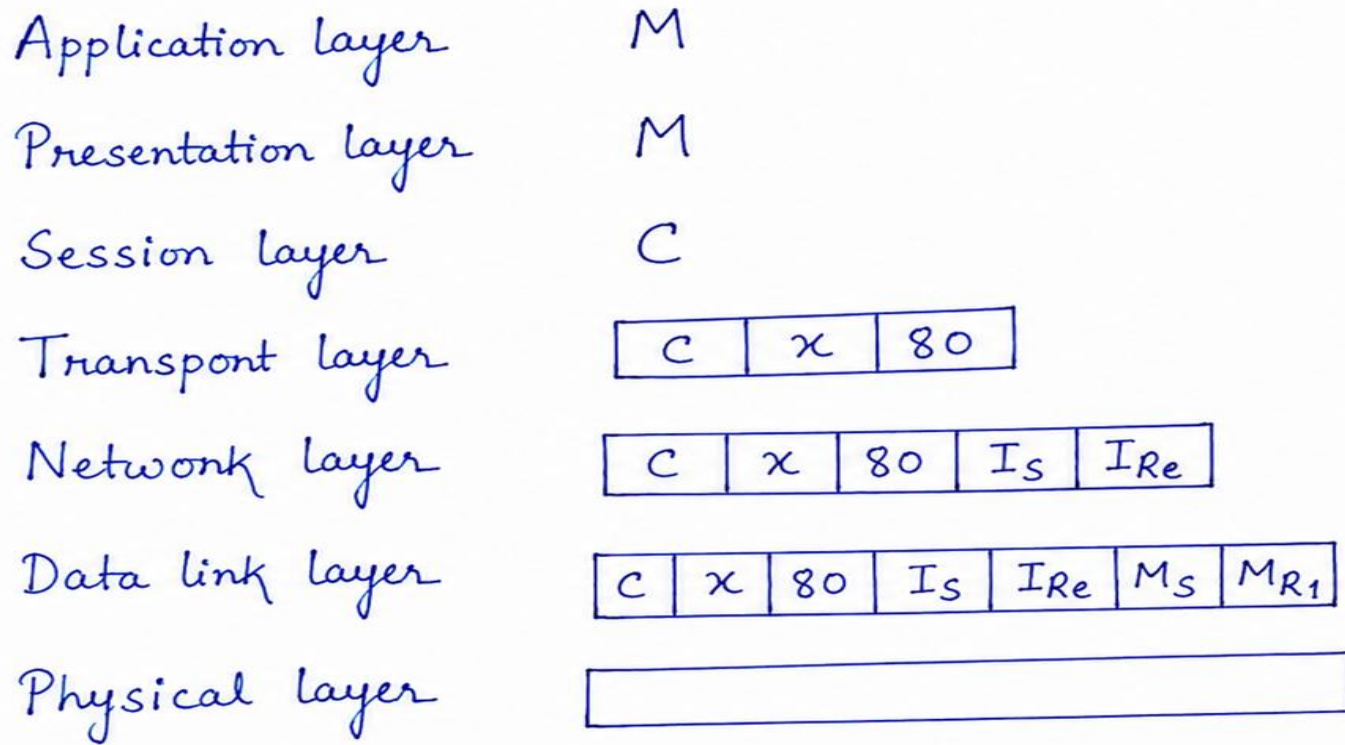
- If Physical link is copper, the message is converted to electrical signal
- If Physical link is optical fiber, the message is converted to lights.
- If it is wireless, it is converted to electromagnetic wave.
- It also deals with topology (How the computers are connected)
  - `- *Bus, Star, Ring, Hybrid*
- Encoding : Given, 1 and 0, how we are going to convert them and send them.



# How OSI Layer works

DNS / Domain name to IP  
ARP - IP → MAC





If R<sub>2</sub> is compromised, what are informations R<sub>2</sub> will have?

- 1) Mac address of R<sub>1</sub>
- 2) IP address of S, R<sub>e</sub>
- 3) Port nos (Source and Dest)
- 4) Ciphertext<sub>C</sub>